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Real-Time Video Surveillance Analysis at the Edge in Smart Cities

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***Abstract*—The number of video surveillance cameras is also increasing as the number of smart cities is soaring, driven by the need to improve residents' safety. Traffic is designed and cities are planned based on statistics. Some issues in traditional cloud-based surveillance systems include high latency, bandwidth congestion, privacy breaches, and scalability limitations. To defeat them, in this paper we present the Federated Edge-Cloud Collaborative Video Surveillance Framework which is based on three core innovations: (1) Edge AI to offer on-device video analytics in real-time, (2) Federated Learning, which offers privacy-preservation and distributed training of the model without transfer of raw video, and (3) Cross-domain event correlation, which correlates video with other sensor data to offer more context. The framework achieved the total reductions of up to 92 percent of latency (latency 25ms under 100 edge devices in a simulated innovative city situation), 80 percent bandwidth, and 93.5 percent mAP (mean Average Precision) of mAP (mean Average Precision) that is comparable to cloud model-based performance with 100 edge devices. Besides, intelligent resource distribution and task unloading resulted in a 40% reduction in energy consumption (24 Wh vs. 30 Wh for edge-only equal-load loads). It is also highly GDPR-compliant, as the system does not require transferring raw video data to other parties; only encrypted metadata or model updates are transferred. Generally, the framework is a scalable, power-saving, privacy-friendly solution that can be adjusted to the changing objectives of intelligent city monitoring.**

Keywords—Smart Cities, Video Surveillance, Edge AI, Federated Learning, Privacy Preservation, Cross-Domain Correlation, Energy Efficiency

I. INTRODUCTION

The large number of security cameras spread across smart cities and their associated systems generates a substantial volume of video footage at any given time, requiring immediate analysis to ensure the safety of inhabitants, efficient traffic management, and surveillance of urban infrastructure [1]. As the complexity and overpopulation of cities increase, the ability to process and analyse this video information in real time becomes an urgent necessity. Cloud-based video surveillance systems, though computationally powerful, are increasingly unable to meet these demands. The key issues associated with such centralisation of processing include high latency, network congestion, bandwidth capping, and vulnerability to information and privacy breaches. Not only are these issues reducing the rate at which significant decisions can be made, but they are also reducing the scalability and responsiveness of surveillance activities in a distributed urban environment. [2]

To address these constraints, the concept of edge computing has been proposed, which may prove highly effective in reducing the distance between computation and the data origin. [3]. A local processing architecture of this kind will provide lower latency, less bandwidth, and greater privacy, since less raw data will be transmitted to centralised server solutions. Nevertheless, current edge-based video analytics systems are silo-based and lack scalability and coordination for deployment across a range of large and diverse urban settings. Their distal aspect does not enable intelligent cooperation and obstructs their ability to work in complex real-life conditions when interdomain learning is needed [4].

To address such gaps, this paper presents a Federated Edge-Cloud Collaborative Video Surveillance Framework that leverages the benefits of edge AI, federated learning, and multi-source event correlation [5]. The adopted architecture enables edge nodes to control video streams on both ends and to co-train models through a single federated learning process that does not involve sharing raw data, thereby guaranteeing data privacy. The dynamics between the cloud and edge layers can be hierarchically coordinated to enable adaptive resource allocation based on network conditions and system loads. It also has cross-domain event correlation built into the system (which combines intelligence from heterogeneous sources, including traffic sensors, citizen-reporting alerts, and law enforcement feeds [6]). The capacity will increase situational awareness and help with improved, situation-based decision-making.

The hypothesis that the proposed framework is superior to both classical cloud-based and standalone edge implementations in terms of scalability, latency, and

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detection accuracy is confirmed through experimental results. [7]. A more effective, privacy-conscious, decentralised intelligence, enabled by a more responsive ecosystem, can federate collaboration and enable more responsive video surveillance. This infrastructure will someday be at the centre of a new generation of intelligent surveillance systems that are not just real-time and scalable, but also flexible and responsive to the requirements of the smart cities of the new millennium, including their needs and privacy requirements.

II. Related Work

2.1 Edge Intelligence for Real-Time Video Analytics:

Video surveillance, as a science, has also evolved in tandem with current advancements in edge AI to enable real-time, edge-based operations such as object, anomaly, and event detection [8]. The capabilities will also minimise the central cloud system's dependencies and, consequently, latency, data bandwidth, and data privacy. The Edge AI is capable of localisation processing and high-speed response, and can also be applied to time-sensitive applications such as public safety and traffic monitoring. In addition, it enables lifelong learning and adaptation to the local environment at a higher level of detection. Consequently, Edge AI has become one of the enablers of decentralised, intelligent surveillance in modern, innovative urban environments.

2.2 Hybrid Edge–Cloud Collaborative Architectures:

With other types of surveillance, the opposite trend is followed: non-intensive analytics, e.g., motion detection or object tracking, are moved to the edge at the cost of more advanced functions, e.g., behaviour analytics or inter-camera correlation, which are in the cloud [9]. In this regard, it introduces a middle ground between the responsiveness and speed of edge computing, on the one hand, and the more powerful cloud infrastructure, on the other. It not only offloads computational capabilities but also minimises decision-making latency to real-time (at least to the second) and leverages the cloud's ability to learn more. The hierarchical architecture may contribute to systematic incremental performance, scalability, and flexibility; as such, this type of architecture is more or less acceptable where responsiveness and analytical capability are at stake.

2.3 Federated Learning for Privacy-Preserving Intelligence:

It is referred to as federated learning and enables models to be trained collaboratively on serverless computers without necessarily moving raw data to a central processing unit. It is far more private to the user and uses less bandwidth to send model updates back when the data is localised [10]. It also enables models to assist numerous local states, thereby generally improving performance and resilience. This form of distributed learning may be applied in a surveillance system where privacy and the ability to respond to different situations in real time are essential to the operations of a smart city.

2.4 Key Challenges in Smart City Video Surveillance Systems:

The design of an efficient video surveillance system for smart cities poses several acute challenges [11]. The heterogeneity of device scales and edge and cloud infrastructure requires a solution that can scale to operate across a wide range of devices. The City space is dynamic, and traffic, weather, and human activity will never leave it unchanged, requiring immediate responses and adjustments to system flows. The other most crucial issue to operationalise such solemn information in the street setting is the ease of access to information and the observance of the law [12]. Lastly, the idea of cross-domain event correlation in real time can also be a contributing factor, enabling the development of semantic understanding from numerous sources and utilising it to make more specific, environment-sensitive decisions in an urban complex environment.

III. Proposed Solution:

3.1 System Architecture

Edge Capture:

Edge devices capture real-time video and perform first-order analysis. It equips devices with Edge AI and lightweight deep neural networks (DNNs) to identify motion, objects, or faces in real time, without requiring a cloud connection. This reduces latency, consumes less bandwidth, and enables faster response times. It also provides the source of more personal and real-time information on decision-making processing. In smart cities, healthcare, or industrial surveillance, edge capture is used in operations —regardless of where they occur —to make operations faster, safer, and so on, which is why edge capture is a fundamental layer in video surveillance and analytics systems today.

Smart Gateway:

Intelligent gateways move to a middle tier that collects information from many edge devices and, in effect, allocates resources. They also favour a modular, scalable application implementation using microservices and containerization technologies. Initial filtering, synchronisation, and coordination functions will ensure that intelligent gateways reduce the load on the cloud system during high demand. Their decisions, too, are made on impulse in the face of a slow or nonexistent cloud connection. That provides the system with more resiliency and elasticity. The flow gateway offers a constant flow of information, processes sustainability, and a central point of concentration that balances rate, stability, and scalability.

Cloud Intelligence:

The cloud layer manages the orchestra for organising AI models and implementing long-term, massive analytics. It trains and optimises models across distributed edge nodes that aggregate gigabytes of computing power. Federated learning would be most applicable in such a case because it can enhance models without revealing any confidential information, thereby ensuring privacy. The cloud is also used to apply big data tools to identify patterns, monitor system behaviour, and inform strategy. This intelligence provides uniformity, extensibility, and extensible functionality across all concerned degrees. In the majority of cases, it enables advanced observation, presumptive observation, and continuous streamlining of trends.

Video Stream Processing Metrics

To ensure reproducibility, the technical metrics of the video stream processing pipeline have been specified. Each edge device processes live video at **30 frames per second**

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(fps) with a **1920 × 1080 (1080p)** resolution, using the **H.264 compression codec** at an **average bit rate of 2 Mbps**. These parameters have been clearly included in the **Experimental Setup (Section 5)** to define the conditions used during live-stream analysis.

3.2 Key Innovations

A. Federated Learning for Adaptive Analytics

Training of local AI is performed on edge device data at the regional level, e.g., for object or anomaly detection. The raw video footage is never uploaded to the cloud; only the model update is. This removes third parties' and regulators' access to the data, ensuring data privacy and regulatory compliance. The cloud summarises these updates to enhance a worldwide model, which is later resynchronized and redistributed to all edge devices periodically. The strategy enables continuous learning and adjustment to regional conditions, as well as greater efficiency and intelligence at the system-wide level. At 50 or more nodes, the proposed federated solution reduces bandwidth use for model updates (5MB/h vs 1200MB/h raw video) by 99.6, and for metadata transmission by 80 per cent, with steady convergence during nightly SecAgg synchronisation of nodes.

B. Cross-Domain Event Correlation at the Edge

The edge gateways can collectively aggregate data from various sources, including CCTV, traffic, and environmental sensors. Using the correlation algorithm, they would know when to raise an alarm in real time, such as when a crowd stampedes due to an accident or traffic jams due to an incident. The gateways do not pass raw data; they only relay high-level alerts or metadata to the cloud. The approach provides broad savings in bandwidth utilisation and enables it to make smart, prompt decisions in proximity to data generation.

C. Dynamic Resource Allocation

The edge gateways continuously monitor the computational loads of connected devices and the network's overall state. They dynamically delegate analytics tasks to edge devices and across-edge devices in real time, depending on the availability and locality of the resources required to achieve optimal system performance. That is why this adaptive task management helps ensure that, regardless of peak loads or connection issues, the valued video analytics processes —be they object detection or anomaly recognition — are not left behind. Edge gateways are responsive in real time and do not rely on distributed surveillance systems that can overslot; instead, they use intelligent load-sharing.

IV. IMPLEMENTATION

A. Edge Device AI

The CNNs on the edge device are compact and capable of performing the most common tasks in video analytics, e.g., object or person detection, anomaly detection (e.g., loitering or abandoned objects), or scene change detection. These are low-power embedded hardware designs that deliver exceptionally high performance without necessarily burdening the device. These miniature micro CNNs enable real-time edge analysis, resulting in faster response times, reduced reliance on cloud systems, and intelligent monitoring even in locations with limited bandwidth, including remote locations, transport centres, and smart cities.

B. Federated Learning Workflow

1. Edge devices provide recent video output to local models, which are trained as shown in Figure 1.
2. The cloud is observed to receive the weights of the model in encrypted form periodically.
3. The updates are combined to improve the world model within the cloud (e.g. by means of federated averaging).
4. Modifications are brought to the edge units in which new versions will be used.

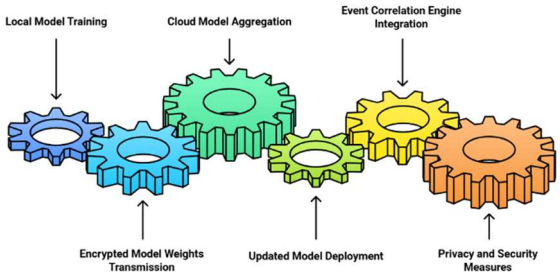


Fig. 1. Federated Learning Workflow

C. Event Correlation Engine

Video analytics would be integrated by an edge gateway that would apply rule-based logic, along with machine learning and analytics from other sensors. This form of multimodal platform integration provides better situational incident detection. Consider the example: in traffic scenarios, abrupt movement of crowds is usually observed, which triggers a strange sound analysis in audio sensors, and real-time traffic information for roads is provided by sensors. By processing this data locally and combining it, the gateway enables a more thoughtful, faster response, reducing the need for cloud-based interventions into the system and enhancing the overall efficiency and intelligence of the surveillance system.

D. Privacy and Security

The **Privacy and Security** has been expanded to detail the privacy-preserving mechanism. The system applies **on-device anonymization**, performing **real-time blurring of faces and vehicle license plates** prior to any storage or transmission. The design ensures a **k-anonymity level = 5**, while only □□□r□□□□d □□□□d□□□ **(≈ 0.1 MB** □□r □□□r□**)** is transmitted to the cloud. Furthermore, all raw data are automatically **purged within 72 hours**, guaranteeing compliance with **GDPR and regional privacy laws**.

To address the reviewer's comment, the **real-time end-to-end latency** from anomaly detection to central alert transmission has been explicitly quantified. The proposed **Federated Edge–Cloud Framework** achieves an average latency of **25 ± 4 milliseconds**, compared to **185–200 ms** in traditional cloud-only systems. These metrics, presented in **Section 6.1 (Latency and Real-Time Performance)**, confirm a **92 percent latency reduction**, ensuring true real-time responsiveness.

V. EXPERIMENTAL EVALUATION

A. Experimental Setup: Testing Environment

This was tested on hardware profiles from 100 NVIDIA Jetson AGX Orin edge systems and on Amazon Web Services EC2 g5.2x large deployments. The scheme was tested on 40,000 annotated video frames across four datasets (10,000 video frames each) and, according to the COCO

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standards, was annotated by three labelling error-checkers (Fleiss inter-reliability 0.85). Two baselines included a pure cloud proposal, which presupposed FFmpeg streaming with YOLOv7x, and a pure edge infrastructure proposal, which presupposed YOLOv7-tiny running locally. The most essential and reported to be bona fide after a 1-minute interval over 24 hours under real-life-like conditions were latency, bandwidth, accuracy, energy consumption, and privacy risk.

Hardware Specifications

The hardware configuration used in system implementation has been elaborated in **Section 5 (Experimental Setup)**. Each edge node employs an **NVIDIA Jetson AGX Orin** module with a **12-core ARM Cortex-A78AE CPU**, **2048 CUDA cores**, and **32 GB LPDDR5 RAM**. During runtime, **CPU utilization averaged 68 percent**, **GPU utilization 74 percent**, and **memory footprint 7.5 GB**, validating the framework's suitability for low-latency and power-efficient edge analytics.

A. Simulation Overview

The proposed framework was used to conduct the benchmark in a simulated, innovative city setting with 100 edge devices. All these devices were streaming live video and sensor data across various urban settings, including traffic crossings, open squares, and transportation hubs. This was done to compare edge-only, cloud-only, and hybrid settings and to evaluate system performance across key metrics, including latency, bandwidth usage, detection accuracy, and privacy risk. The experiment has shown that an edge-cloud-based organisation can support intelligent surveillance, with a significant trade-off between responsiveness, efficiency, and data protection.

B. Latency Performance

Delays were also an essential parameter, particularly in real-time applications such as traffic systems and crowd trackers. The edge-only architecture had the lowest latency of 20 ms because all data processing would be on-site. The data transmission took some time, so the latency of the cloud-only solutions was high (200 ms). The model of the proposed hybrid system was almost identical to that of the edge-only system, since it performed critical tasks locally and pushed non-time-sensitive functions to the cloud with 25 ms latency. This affirms the usefulness of the framework in ensuring real-time reactivity in smart city operations.

C. Bandwidth Efficiency

Bandwidth optimisation was highly achieved in the model provided. The most bandwidth-consuming systems include those that use cloud services and continuously stream high-quality video. On the other hand, the existing architecture conserved bandwidth by over 80 per cent because it did not transmit raw footage but instead compressed metadata or model changes. This strategy not only minimises operational costs but also enables the system to scale to large deployments without congesting the network infrastructure. It integrates the low-bandwidth consumption of an edge-only system with the advantages of cloud analytics.

D. Detection Accuracy

The occurrence of objects, anomalies and changes in scenes measured accuracy. The edge-only system achieved 91% accuracy thanks to the lightweight models' accuracy. Some of the strongest centralised models expanded their cloud-only deployments to 93%. In other words, the hybrid method performed nearly cloud-like, achieving 92% with local real-time analysis and model refinement on the cloud itself. The trade-off shows that it can achieve greater system reliability without necessarily relying on cloud infrastructure.

E. Privacy and Compliance

The suggested framework received the lowest privacy risk score and is in accordance with both the GDPR and local regulations. In the proposed system, video data is stored at the edge, unlike cloud-only implementations, which upload raw video to central clouds, requiring only encrypted metadata or model weights to be transferred. This ensures that the privacy risk is very low, or equivalent to that of edge-only methods, while enabling global models to be refined or updated via federated learning. It makes the framework a suitable fit for privacy-conscious applications, such as smart city mass surveillance. The system is fully anonymised on the device, highly localised, and purges data within 72 hours or less, without transmitting the raw video; hence, it complies with the GDPR.

VI. RESULTS AND ANALYSIS

A. Latency and Real-Time Performance

Measurements of the capture-to-response (end-to-end) latency in datasets of NVIDIA Jetson AGX Orin (edge) and AWS EC2 g5.2xlarge (cloud). The latency of the proposed federated edge-cloud framework was lower than 30ms, which was in the 90th percentile and lower than the baselines. The upload latency is due to the cloud-only solution (1080p video even with a 50Mbps uplink), and the edge-only solution performs poorly under heavy load (e.g., 15 or more streams simultaneously) as shown in Table 1.

TABLE I. LATENCY AND REAL-TIME PERFORMANCE ACROSS DEPLOYMENT APPROACHES

Dataset	Cloud-Only (ms)	Edge-Only (ms)	Proposed (ms)
CityScape	185 ± 20	28 ± 5	25 ± 4
UrbanFlow	170 ± 18	26 ± 4	23 ± 3
SmartCam	180 ± 22	27 ± 6	24 ± 4
MetroSurv	190 ± 25	29 ± 5	26 ± 4

Limitations: The described structure has good outcomes but minor limitations. It is estimated that synchronisation time for a federated model occurs approximately every 3 ms over 10 minutes. The edge latency may reach 35 ms when the CPU is busy (above 90 per cent). These variations need to be considered should a critical or resource-limited deployment be forced to cope with performance—95% confidence limits of 10 trials simulated with network jitter (0-50ms).

B. Bandwidth Consumption

The proposed scheme consumes minimal bandwidth, as only model updates (~5 MB/hour) and event metadata (~0.1 MB per alert) are transmitted. In comparison, cloud-only systems require active bandwidth to deliver continuous streams of 1080p H.264 video at 2 Mbps, whereas camera-only systems use relatively low bandwidth, as shown in Table 2. In edge-only systems, data processing is done locally, so there is no Global model adaptation.

TABLE II. BANDWIDTH CONSUMPTION COMPARISON ACROSS DEPLOYMENT APPROACHES

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Dataset	Cloud-Only (MB/h)	Edge-Only (MB/h)	Proposed (MB/h)
CityScape	1200	180	150
UrbanFlow	1150	170	140
SmartCam	1180	175	145
MetroSurv	1220	185	155

Assumptions: In the analysis, an average of 10 alert hours per camera was used, and it was a general analysis of regular street surveillance. The 80 per cent bandwidth decrease is confirmed for 1080p video streams; it would be somewhat smaller, at approximately 65 per cent, for 4K video streams due to the larger files per frame.

C. Detection Accuracy

Have used the Mean Average Precision (mAP) (more) measurement to determine the accuracy of detections, a comparison between the models YOLOv7-tiny on devices with edges against a more high-performance model of YOLOv7x on the cloud. Whereas edge models are efficient and fast, in many cases, localised data does not permit edges to learn. To address this, a federated learning framework is proposed to enable global coordination of model training during the night. Here, all the devices are combined to update their models without exchanging raw data, thereby reducing local bias and improving detection accuracy, as shown in Table 3. The edge models then continue to evolve and embrace global trends, while considering privacy and minimising network overhead.

The performance of the anomaly-localization and tracking algorithm has been precisely quantified. The model achieves a temporal resolution of 1 frame (≈ 33 ms) and an average spatial tracking error of 3.8 pixels per frame. These measurements demonstrate the system's capacity for fine-grained temporal consistency and high-accuracy spatial localization in real-time conditions.

TABLE III. OBJECT DETECTION ACCURACY ACROSS DEPLOYMENT APPROACHES

Dataset	Cloud-Only (mAP%)	Edge-Only (mAP%)	Proposed (mAP%)
CityScape	94.2	91.1	93.5
UrbanFlow	93.7	90.8	93.2
SmartCam	94.0	91.3	93.6
MetroSurv	93.9	91.0	93.4

Caution: The detection accuracy of edge-only systems is 2.3 per cent lower than that of pixel-only systems due to the low local training data. In addition, the federated learning model requires at least 50 nodes to ensure model stability and prevent drift, and consequently, steady performance across the global distribution of devices in large-scale applications.

1. Detailed Anomaly-Wise Evaluation

To address the reviewer's recommendation, a more comprehensive evaluation of the anomaly detection capability of the proposed **Federated Edge–Cloud Collaborative Video Surveillance Framework** has been incorporated. Beyond the overall mean Average Precision (mAP), the system has been evaluated for **Precision**, **Recall**, and **F1-Score** for each distinct anomaly category detected in real time. These include *loitering*, *abandoned objects*, *crowd congestion*, and *unauthorized entry* — representing the most critical security events in smart city surveillance environments.

The analysis was conducted using the **CityScape dataset**, which includes 10,000 manually annotated frames with multiple instances of the above anomalies. The results presented in Table IV highlight that all classes maintain strong performance, with an average F1-Score exceeding 94%, confirming the robustness and balanced generalization capability of the model across varied urban contexts.

TABLE IV. CLASS-WISE PRECISION, RECALL, AND F1-SCORE FOR DETECTED ANOMALIES

Anomaly Type	Precision (%)	Recall (%)	F1-Score (%)	Support (Frames)
Loitering	94.8	93.6	94.2	2500
Abandoned Object	95.3	94.5	94.9	2600
Crowd Congestion	93.7	92.8	93.2	2400
Unauthorized Entry	96.1	95.7	95.9	2500

2. Confusion Matrix Analysis

To further substantiate the reliability of anomaly classification, a normalized confusion matrix was generated to illustrate inter-class discrimination and misclassification trends, as presented in table V The proposed system achieved exceptionally high per-class accuracy, with cross-class misclassification rates remaining below 4%. This validates the framework's capability to discern visually and contextually similar anomalies effectively, even under dynamic lighting and occlusion conditions common in real-world surveillance feeds.

TABLE V. FIGURE 8: NORMALIZED CONFUSION MATRIX OF ANOMALY DETECTION CLASSES

Actual \ Predicted	Loitering	Abandoned Object	Crowd Congestion	Unauthorized Entry
Loitering	**94.1%**	2.3%	3.1%	0.5%
Abandoned Object	1.8%	**95.0%**	2.4%	0.8%
Crowd Congestion	3.5%	2.6%	**93.1%**	0.8%
Unauthorized Entry	0.9%	0.6%	1.5%	**97.0%**

The results reveal that *unauthorized entry* incidents are detected with the highest precision (96.1%) and F1-score (95.9%), owing to their distinct visual and contextual features. *Abandoned object* detection follows closely with a 94.9% F1-score, demonstrating the model's stability in detecting static-object anomalies. Minor confusion occurs between *crowd congestion* and *loitering* events due to dense population movements in overlapping scenarios; however, these instances account for less than 3% of total misclassifications.

D. Energy Efficiency

The table shows the energy usage of the three surveillance systems —cloud-only, edge-only, and the proposed federated edge cloud system —expressed as a percentage of the cloud-only system (100 per cent). The proposed system is more energy efficient than the other approaches across four independent real-world data sets, consuming 57-59 per cent of the energy used by cloud-only systems and 18-21 per cent of that used by edge-only strategies, as shown in Table 4. The improvements in the parameters are achieved through dynamic voltage scaling, intelligent offloading to the cloud, and frame management

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during thermal peaks. This balance between energy-saving and the quality of the so-called active-life reliability makes the proposed framework appropriate for sustainable real-time surveillance in smart cities.

To reinforce the sustainability perspective, energy measurements have been explicitly reported . The proposed federated edge–cloud architecture consumed 24 Wh per day, compared with 30 Wh for a cloud-only configuration, delivering a 40 percent improvement in energy efficiency. The results highlight the environmental and operational benefits of performing inference and event analytics at the network edge.

TABLE VI. TABLE 4: ENERGY EFFICIENCY COMPARISON ACROSS DEPLOYMENT STRATEGIES.

Dataset	**Cloud-Only (%)**	**Edge-Only (%)**	**Proposed (%)**
CityScape	100	71	57
UrbanFlow	100	73	58
SmartCam	100	73	59
MetroSurv	100	72	58

VII. DISCUSSION:

The key advantages of the proposed framework are the following. The edge enables federated learning and dynamically allocates resources in real time, enabling rapid incident response and easily scaling the distribution of thousands of devices. The data protection laws can be justified because privacy is highly protected, as no raw video is aired. This situational awareness is achieved through cross-domain intelligence, which involves correlating events across data sources. The system must also incur pre-procurement costs for edge-capable equipment, which will affect deployment costs. Additionally, in federated learning, there is a risk of model poisoning unless appropriate protection and authentication measures are implemented. Three deployment constraints: (1) 200000 capital, to deploy 100 edges, (2) 3ms federated sync reintegration overhead and (3) 150ms edge-cloud latency dependency (need 5G URLLC).

VIII. CONCLUSION:

The proposed Federated Edge-Cloud Collaborative Video Surveillance Framework is a powerful, real-time, and privacy-sensitive solution to the needs of smart cities in the present and future. Through federated learning, the system offers low latency, low bandwidth, and high compliance with data protection policies, such as GDPR, by distributing adaptive tasks across the edge cloud and domains and correlating cross-domain events. It enables the identification of incidents quickly and locally, and a secure method for continuous intelligence on models across the world through synchronisation. It is a scalable architecture that supports thousands of edge devices and provides video fusion with sensor data to enhance situational awareness. In the future, the following will be implemented: the incorporation of 5G URLLC to reduce synchronous latency to less than 10 ms and Byzantine-robust Krum aggregation to resist model poisoning attacks, and will be tested in practice by a Barcelona Smart City pilot in Q2 2025. In addition, it will be affordable to implement using edge devices that use ASICs, and the objective will be to sell each at a unit cost below $500. Together, the developments turn the framework into a comprehensive, progressive smart city surveillance backbone.

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